

**Client Change Request (CCR)****CCR Number:** CCR-005**Client:** Downtown Parking Services**Submitted By:** Robert Williams, Operations Manager**Priority:** High**Requested Delivery:** End of Today's Sprint**Assigned Developer:** \_\_\_\_\_**Date Assigned:** \_\_\_\_\_

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**Business Background**

Downtown Parking Services operates a parking garage in the city center that serves commuters, shoppers, and visitors throughout the day. As vehicles leave the garage, employees calculate parking charges based on the amount of time each vehicle has been parked.

Currently, attendants manually calculate parking fees and record the day's totals using paper forms. This process is slow, increases the likelihood of calculation errors, and makes it difficult to determine how much revenue the garage generated during the day.

The parking garage manager would like a simple computer program that allows employees to continue entering parking information for each vehicle until all vehicles have been processed. When the work is complete, the program should provide a summary of the day's activity.

At this time, the manager is only interested in processing one day's parking activity.

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## **Customer Requirements**

The employee should be able to enter the amount of time that each vehicle was parked.

After each parking time is entered, the program should calculate the parking charge for that vehicle and display it to the employee.

The employee should continue entering vehicles until they indicate that there are no more vehicles to process.

When all vehicles have been processed, the program should display:

- The total number of vehicles processed.
- The total revenue collected.
- The average parking fee charged per vehicle.

The report should be clearly formatted and easy for employees to read.

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## **Acceptance Criteria**

The customer will consider the software complete when all of the following are true.

- ☐ Employee can enter parking information for multiple vehicles.
  - ☐ Program calculates and displays the parking charge for each vehicle.
  - ☐ Employee can indicate when there are no additional vehicles to process.
  - ☐ Program correctly counts the total number of vehicles processed.
  - ☐ Program correctly calculates the total revenue collected.
  - ☐ Program correctly calculates the average parking fee.
  - ☐ Program displays a neatly formatted summary report.
  - ☐ Program terminates normally after the summary has been displayed.
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## Out of Scope

The customer is **not** requesting:

- Saving parking information to a file
- Printing receipts
- Processing multiple parking garages
- Assigning parking spaces
- Accepting customer names or license plate numbers
- Credit card or cash payment processing
- Monthly parking permits
- Taxes or discounts
- Displaying charts or graphs

If these features appear in the software, they will not be evaluated.

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## **Example Scenario 1**

### **Employee Input**

Vehicle 1 Hours Parked: 2

Vehicle 2 Hours Parked: 5

Vehicle 3 Hours Parked: 3

More vehicles? Y

Vehicle 4 Hours Parked: 7

More vehicles? N

### **Expected Output**

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Downtown Parking Services

Parking Summary

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Vehicle 1 Charge : \$4.00

Vehicle 2 Charge : \$10.00

Vehicle 3 Charge : \$6.00

Vehicle 4 Charge : \$14.00

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Vehicles Processed : 4

Total Revenue : \$34.00

Average Parking Fee: \$8.50

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## Example Scenario 2

### Employee Input

Vehicle 1 Hours Parked: 1

More vehicles? N

### Expected Output

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Downtown Parking Services

Parking Summary

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Vehicle 1 Charge : \$2.00

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Vehicles Processed : 1

Total Revenue : \$2.00

Average Parking Fee: \$2.00

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### **Notes from the Customer**

"Our attendants should not have to restart the program after every vehicle. They should simply continue entering parking information until the garage is empty. At the end of the day, I want a summary showing how many vehicles were processed, how much money was collected, and the average amount each customer paid."

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### **Developer Notes**

This section intentionally left blank.

Students will complete the software requirements during class before beginning implementation.

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### **Questions for the Customer**

Before writing any code, list any questions you would ask the customer to clarify the request.

Examples:

- 1.

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## Reminder

The customer has intentionally described the problem using **business language**, not programming terminology. During class, your first task is to translate these customer requirements into **software requirements** before writing any C code.

For this assignment, you should use only the C language features that have been covered through **Lesson 4**, including:

- Variables and constants
- Input/output
- Selection statements (if and switch)
- Loops (while and for)
- break and continue
- Basic arithmetic and formatted output

Do **not** use C language features that have not yet been introduced in the course.